

"Osprey 3.5 V6 high-pressure fuel pump failure causing P0087 low rail pressure"

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Area: bulletins

1 Condition

The Osprey 3.5 V6 may set DTC P0087 (Fuel Rail/System Pressure Too Low) with symptoms including extended cranking, hard starts when hot, power loss under heavy throttle, and an illuminated MIL. The PCM commands rail pressure that the system cannot achieve, triggering P0087 and possible limp-home operation. Verify the condition by comparing commanded versus actual rail pressure on a scan tool before any parts replacement. Use the Osprey Fuel Service Manual :: Fuel System Diagnosis to confirm the low-pressure condition is real and not a sensor artifact.

2 Cause

P0087 on this engine is most often caused by mechanical wear in the high-pressure fuel pump, part number HPFP-4120, where the pump plunger or check valve fails to build commanded GDI rail pressure. A secondary contributor is a drifting fuel rail pressure sensor, part number FRS-4180, which can report falsely low pressure and set the same code. Confirm whether HPFP-4120 is failing to deliver pressure or whether FRS-4180 is misreporting before condemning the pump. The Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults and the Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) describe the pinpoint tests.

3 Repair Procedure

When testing confirms a failed pump, replace HPFP-4120 following PROC-FUEL-PUMP, observing all GDI high-pressure depressurization and cleanliness precautions. If diagnosis instead points to a faulty sensor, replace FRS-4180 and clear P0087 before retesting rail pressure. Always use a new cam-follower and tighten fittings to specification to avoid leaks. Detailed steps are in the Osprey Fuel Service Manual :: Fuel Pump Replacement Procedure and the Osprey Fuel Service Manual :: Fuel Rail Pressure Sensor Replacement . For broader system context consult the Osprey Fuel Service Manual :: Fuel Delivery section.